Before entering the main program, the following constant and variables must be defined:

NR_COEF	The number of coefficients in the filter kernel (301 in this example)
dline[NR_COEF]	A circular buffer holding the past input samples, in data memory
coef[NR_COEF]	A circular buffer holding the filter coefficients, in program memory

```
001  /************************************************************************
002  ******************        MAIN PROGRAM        **********************
003  ************************************************************************/
004  main:
005
006  /* INITIALIZE THE DAGS TO CONTROL THE CIRCULAR BUFFERS */
007
008  b0 = dline;          /* set up dline[ ], the buffer holding the past input samples */
009  l0 = @dline;
010  m0 = 1;
011  b8 = coef;           /* set up coef[ ], the buffer holding the filter coefficients */
012  l8 = @coef;
013  m8 = 1;
014
015  /* ENTER A LOOP, WAITING FOR THE SAMPLE-READY INTERRUPT */
016
017  wait:
018  idle;
019  jump wait;
020
021
022  /************************************************************************
023  *********     SUBROUTINE TO PROCESS ONE SAMPLE        **********
024  ************************************************************************/
025  sample_ready:
026
027  /* ACQUIRE THE INPUT SAMPLE, CONVERT TO FLOATING POINT */
028
029  r0 = dm(rx_buf + 1);          /* move the input sample into r0 */
030  r0 = lshift r0 by 16;         /* shift to the highest 16 bits to preserve the sign */
031  r1 = -31;                     /* set the scaling for the conversion */
032  f0 = float r0 by r1;          /* convert from fixed to floating point */
033  dm(i0,m0) = f0;               /* store the new sample in dline[ ], and zero f12 */
034
035  /* CALCULATE THE OUTPUT SAMPLE FROM THE FIR FILTER */
036
037  f12 = 0;                                        /* prime the registers */
038  f2 = dm(i0,m0), f4 = pm(i8,m8);
039  f8 = f2*f4, f2 = dm(i0,m0), f4 = pm(i8,m8);
040                                                  /* efficient main loop */
041  lcntr = NR_COEF-2, do (pc,1) until lce;
042  f8 = f2*f4, f12 = f8+f12, f2 = dm(i0,m0), f4 = pm(i8,m8);
043
044  f8 = f2*f4, f12 = f8+f12;                       /* complete the last loop */
045  f12 = f8+f12;
046
047  /* CONVERT THE OUTPUT SAMPLE TO FIXED POINT & OUTPUT */
048
049  r1 = 31;                    /* set the scaling for the conversion */
050  r8 = fix f12 by r1;         /* convert from floating to fixed point */
051  rti(db);                    /* return from interrupt, but execute next 2 lines */
052  r8 = lshift r8 by -16;      /* shift to the lowest 16 bits  */
053  dm(tx_buf + 1) = r8;        /* move the sample to the output */
```

TABLE 29-2
FIR filter program in assembly.